

# MEDIAN FROM FREQUENCY TABLES

## Foundation Tier

Median, median class and interpretation

Name .....

Date .....

8 questions

9 marks

2017-2025

### Instructions

- Answer each question in the space provided.
- Show all your working and follow the rounding instructions.
- Diagrams are not accurately drawn unless stated otherwise.
- Use a calculator except for pages marked NON-CALCULATOR.

### Using this booklet

Table questions are arranged by exam series, followed by any chart or histogram extensions. Each page gives its original paper reference and question number. The question numbers within the extracted artwork are the original exam numbers.

Only the specified question parts are included. Shared questions appear in both tier booklets so that each can be used independently. Original question numbers and part labels are retained.

This is an independently compiled practice resource, not an official Pearson examination paper. The source collection covers standard UK June and November papers from 2017-2025. No solutions are included.

# Question index

Foundation Tier | 8 questions | 9 marks

| Practice | Exam series   | Paper   | Original question | Marks | Page |
|----------|---------------|---------|-------------------|-------|------|
| 01       | June 2017     | 1MA1/3F | Q17(a)            | 1     | 3    |
| 02       | November 2017 | 1MA1/3F | Q19(a)            | 1     | 4    |
| 03       | June 2019     | 1MA1/3F | Q26(a)            | 1     | 5    |
| 04       | November 2020 | 1MA1/1F | Q16(a)            | 1     | 6    |
| 05       | June 2022     | 1MA1/2F | Q15(b)            | 2     | 7    |
| 06       | November 2023 | 1MA1/3F | Q22(b)            | 1     | 8    |
| 07       | June 2024     | 1MA1/2F | Q28(a)            | 1     | 9    |
| 08       | June 2025     | 1MA1/3F | Q19(a)            | 1     | 10   |

The original exam numbering is retained in each question.

Paper 1 questions are marked NON-CALCULATOR on their pages.

17 The table shows some information about the dress sizes of 25 women.

| Dress size | Number of women |
|------------|-----------------|
| 8          | 2               |
| 10         | 9               |
| 12         | 8               |
| 14         | 6               |

(a) Find the median dress size.

(1)

Additional working space

| Height ( $h$ cm)   | Frequency |
|--------------------|-----------|
| $130 < h \leq 140$ | 4         |
| $140 < h \leq 150$ | 11        |
| $150 < h \leq 160$ | 24        |
| $160 < h \leq 170$ | 22        |
| $170 < h \leq 180$ | 19        |

- (a) Find the class interval that contains the median.

(1)

| Height ( $h$ cm) | Frequency |
|------------------|-----------|
| $10 < h \leq 20$ | 7         |
| $20 < h \leq 30$ | 13        |
| $30 < h \leq 40$ | 14        |
| $40 < h \leq 50$ | 12        |
| $50 < h \leq 60$ | 16        |
| $60 < h \leq 70$ | 18        |

- (a) Find the class interval that contains the median.

(1)

**16** The table gives information about the number of points scored by each of 16 students in a game.

| Number of points | Frequency |
|------------------|-----------|
| 0                | 1         |
| 1                | 3         |
| 2                | 5         |
| 3                | 4         |
| 4                | 3         |

Tina worked out the median of the number of points scored to be 5

(a) Explain why it is **not** possible for the median to be 5

(1)

### Additional working space

- 15** The table shows information about the number of social media accounts used by each of 300 students.

| Number of social media accounts | Frequency |
|---------------------------------|-----------|
| 0                               | 3         |
| 1                               | 57        |
| 2                               | 84        |
| 3                               | 75        |
| 4                               | 81        |

- (b) Find the median number of social media accounts used by these students.

(2)

Additional working space

22 Seija works at a weather station.  
The table gives information about the temperature,  $T$  °C, at midday for each of 50 cities in the UK on Tuesday.

| Temperature ( $T$ °C) | Frequency |
|-----------------------|-----------|
| $10 < T \leq 15$      | 2         |
| $15 < T \leq 20$      | 8         |
| $20 < T \leq 25$      | 13        |
| $25 < T \leq 30$      | 21        |
| $30 < T \leq 35$      | 6         |

Seija says,  
“The median temperature is 22.5 °C because 22.5 is the middle number in the middle group.”

(b) Is Seija correct?  
Give a reason for your answer.

(1)

Additional working space

**28** The table shows information about the weights of 120 oranges.

| Weight ( $w$ grams) | Frequency |
|---------------------|-----------|
| $50 < w \leq 100$   | 34        |
| $100 < w \leq 150$  | 29        |
| $150 < w \leq 200$  | 27        |
| $200 < w \leq 250$  | 19        |
| $250 < w \leq 300$  | 11        |

(a) Find the class interval that contains the median.

(1)

Additional working space

**19** Katya counted the number of people in each of 50 cars.

The table gives information about her results.

| Number of people | Frequency |
|------------------|-----------|
| 1                | 16        |
| 2                | 15        |
| 3                | 7         |
| 4                | 9         |
| 5                | 3         |

(a) Find the median number of people.

(1)

### Additional working space